Task list for mathematical analysis

Task 1: Suppose that the sequence $(a_n)_{n\in\mathbb{N}}$, with $a_n \in \mathbb{R}$, converges to a . Show that the sequence $b_n = \frac{1}{n}(a_1 + ... + a_n)$ converges to a as well. Is the converse also true, i.e. does the convergence of (b_n) imply the convergence of $(\dot{a}_n)$?

Task 2: Let $\alpha > 0$. Show that the sequence $(a_n)_{n=1}^{\infty}$ with $a_n = \left(\frac{1}{n}\right)^{\alpha} \log \frac{1}{n}$ converges to zero.

Task 3: Does there exist an uncountable collection $\mathcal{C}$ of subsets of $\mathbb{N}$ such that for any $A, B \in \mathcal{C}$ either $A \subseteq B$ or $B \subseteq A$?

Task 4: Show that the set of all finite subsets of $\mathbb{N}$ is countable.

Task 5: Show that the series $\sum\limits_{n\geq 1}^{\infty} \left(n \cdot \log n \cdot \log\log^{c} n\right)^{-1}$ converges for $c > 1$. What happens for $c = 1$?

Task 6: Show that in order that a set X is infinite, the following condition is necessary and sufficient: for every function f of X into itself there exists a nonempty subset $A \neq X$ such that $f(A) \subset A$.

Task 7 : Describe the set of limit points of the sequence (q_n) with $a_{n+1} = \left(a_n + \sqrt{2}\right) \bmod 1$ and $a_1 = 1$. What can be said if $\sqrt{2}$ is replaced by a rational number $x = \frac{p}{q}$?

Task 8: Let $(c_n)_{n=1}^{\infty}$ be a sequence of positive real numbers . Show that $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} c_n^{1/n}$.

Task 9 : give an example of a divergent series $\sum\limits_{n=1}^{\infty} a_n$ such that $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} a_n^{1/n} = 1$ and give an example of a convergent series $\sum\limits_{n=1}^{\infty} b_n$ such that $\lim_{n\to\infty} \frac{b_{n+1}}{b_n} = \lim_{n\to\infty} b_n^{1/n} = 1$.

Task 10: Let f be a function which assigns to each $x \in [1,2] \subset \mathbb{R}$ an open interval $I_x = (a, b)$ with $b > a > 0$ and $x = \frac{b+a}{2} \in [1,2]$ (note that a, b might be different for each x). Show that it is always possible to choose a finite subset $X \subset \mathbb{R}$ such that $\bigcup\limits_{x\in X} I_x \supset [1,2]$.